

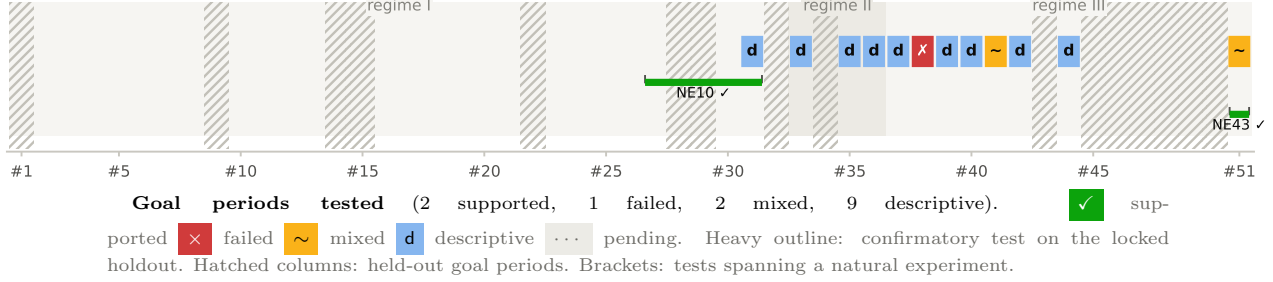
H35: The nudger is a measurably inefficient Maxwell demon

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2/5

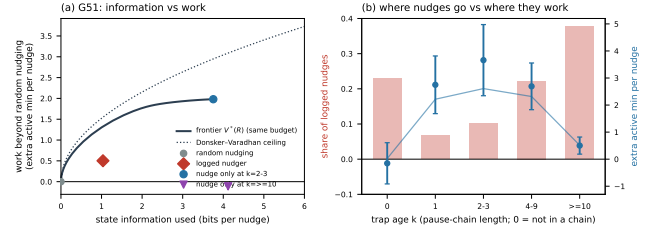
Bottom line. The nudger is a measurably inefficient Maxwell demon for activity (once-early nudging buys 2-3× more active minutes) and a worthless one for work: on the git ledger a nudge buys glances, not commits (+0.22/h, n.s.).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The auto-nudger was treated as a feedback controller (a Maxwell demon). I measured how much agent-state information its nudges carry, how much activity they buy, and how close it comes to the best use of those bits, on 13 non-holdout periods plus two natural experiments (dated predictions; synthetic recovery checked first). In G51, the only well-powered period, a first nudge buys 1.45 [0.72, 2.20] extra active minutes using 1.4 bits of state. That is 38% of the work those bits could buy ($\eta_{SU} = 0.38 [-0.08, 0.68]$); only 15% of the bits were needed. Among pausing agents it does worse than random: 38% of nudges go to traps ≥ 10 pauses deep (0.5 min each, vs 2.2–2.6 at $k = 1-9$). Nudges add 0.5% of active minutes, invisible at switch-on (NE10) and at a silent stop on 08-20. Under the old 12-h pause default, nudges wake agents at any trap age. **Round 1b** (work ledger, ledger calls): the nudger buys glances, not committed work or sustained runs. Holdout: not run.



Observables. (a) G51 in trap-age space: work beyond random nudging against state information used. The curve is the best any policy can do with that many bits (dotted: analytic ceiling). The logged nudger sits far below it; nudging only at $k = 2-3$ reaches the curve's end, and nudging only deep traps is worse than random. (b) The nudger's mass (bars) sits where the response per nudge (points) is lowest.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Directly actionable, though the stakes are small. For an operator running an idle-nudger on LLM agents: nudge each pause chain once, early (the first or second re-pause), and only while the agent is paused. Stop re-nudging agents stuck for many pauses; they barely respond and need something else (a reset, a new task). In G51 that would double or triple the work per nudge at the same budget. The catch: the nudger buys only about 0.5% of agent activity, so even a perfect policy moves the swarm little unless you nudge far more. The thermodynamic framing works as accounting (frontier, ceiling, efficiency), not as physics. The policy result rests on one short-pause period and is untested on the holdout.

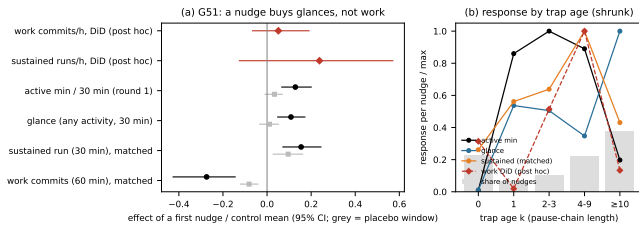
PHYSICAL MODEL

Models 04 (Kolchinsky–Wolpert) and 02. System X : an agent's state (idle time, pause timing, trap age $k = \text{pause-chain length}$). Demon: nudge $M \in \{0, 1\}$ at rate r . Work $g(x)$: extra active minutes in 30 min caused by a nudge in state x . Value of information ΔV : work beyond random nudging. Frontier $V^*(R)$: the most work any policy using R bits can buy. s : half-range of g (plays $k_B T$). p_k : probability of acting at the pause expiry.

$$\begin{aligned} \Delta V(\pi) &= \mathbb{E}[\pi g] - r \mathbb{E}[g], & I &= I(X; M) \\ V^*(R) &= \max_{I \leq R} \Delta V \Rightarrow \pi_\beta = \sigma(\logit r + \beta(g - \lambda)) \\ \eta_{SU} &= \Delta V_{\log} / V^*(I_{\log}), & \eta_{KW} &= R^*(\Delta V_{\log}) / I_{\log} \\ \Delta V &\leq s\sqrt{2rI} \quad (\text{Donsker-Varadhan}) \\ \logit p_k &= \alpha_i + \theta \ln k + (\gamma + \lambda \ln k) \mathbb{I}[\text{nudge}] \end{aligned}$$

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (G51)	Verdict
P1 3–8 bits	1.42 bits/nudge (unchanged)	partly
P2 work 0.5–3 min	1.45 active min → glance +0.08; commits +0.22/h [−0.29, 0.82] (DiD)	min yes; work no
P3 falls with trap age	minutes yes; sustained flat	minutes only
P4 $\eta_{SU} \leq 0.5$	0.38 (min) → glance 0.88; work undefined	activity only
P5 once early $\geq 1.25\times$	$\times 2.3$ min; work $\times 1.3$ (CI spans 0)	min only
NE43 (native)	work -0.03 / -0.49 day-SD at the two steps	yes
NE10 (native)	work $z = +0.92$	yes
G38 vs G51 (native)	sustained 0.058 vs 0.094	no



Second figure. Round 1b, G51. (a) Effect of a first nudge relative to the control mean (grey: placebo window before the nudge). Glances and active minutes rise with clean placebos; sustained runs and work commits fail the placebo (selection: nudged agents had just finished a burst), and after differencing (red) both are zero. (b) Response by trap age: active minutes peak at $k = 2-9$, glances are highest in deep traps, sustained work is flat.

GOAL PERIODS IN DETAIL

G31 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G33 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G35 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G36 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G37 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G38 $\times$ 1b: failed (native N1 as worded: sustained ATT 0.058 ...
G39 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G40 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G41 $\sim$ 1b: mixed (round-1 statistics unchanged; no strictly ...
G42 d 1b: descriptive (i 6 strictly isolated first nudges; ...
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only G51 is powered. The past-only design fails its placebo for sustained runs and work commits, so those numbers rest on a post hoc difference-in-differences. The once-early policy gain is in active minutes; for commits it is not identified. Identification assumes the nudger acts on what we see; agent identity carries extra information. The 08-20 stop is undocumented. Three round-1 amendments and one round-1b post hoc step are disclosed on the card.

ROUND 2

Re-freeze `confirm.py` with glance and work outcomes, then run it on #47 and #50 (C2–C4) and #45 (wake-up). Test the once-early rule as an actual intervention if operators allow. Add an LLM-state feature (Jev behavior labels) to see what the nudger reads beyond trap age.

No round-2 redirects yet.

H35-nudger-maxwell-demon · generated 2026-10-04 11:10 by infra/summaries/build_summaries.py · card hypotheses/H35-nudger-maxwell-demon/README.md · ratings: H35 round-1b agent (suggested) (page 2)